



Vidyavardhini's

College of Engineering & Technology

Vasai Road (W)

Department of Computer Engineering

Laboratory Manual

(Student Copy)

Semester	V	Class	T.E
Course Code	CSL501		
Course Name	Software Engineering Lab		



Vidyavardhini's College of Engineering & Technology

Vision

To be a premier institution of technical education; always aiming at becoming a valuable resource for industry and society.

Mission

- To provide technologically inspiring environment for learning.
- To promote creativity, innovation and professional activities.
- To inculcate ethical and moral values.
- To cater personal, professional and societal needs through quality education.



Department Vision:

To evolve as a center of excellence in the field of Computer Engineering to cater to industrial and societal needs.

Department Mission:

- To provide quality technical education with the aid of modern resources.
- Inculcate creative thinking through innovative ideas and project development.
- To encourage life-long learning, leadership skills, entrepreneurship skills with ethical & moral values.

Program Education Objectives (PEOs):

PEO1: To facilitate learners with a sound foundation in the mathematical, scientific and engineering fundamentals to accomplish professional excellence and succeed in higher studies in Computer Engineering domain

PEO2: To enable learners to use modern tools effectively to solve real-life problems in the field of Computer Engineering.

PEO3: To equip learners with extensive education necessary to understand the impact of computer technology in a global and social context.

PEO4: To inculcate professional and ethical attitude, leadership qualities, commitment to societal responsibilities and prepare the learners for life-long learning to build up a successful career in Computer Engineering.

Program Specific Outcomes (PSOs):

PSO1: Analyze problems and design applications of database, networking, security, web technology, cloud computing, machine learning using mathematical skills, and computational tools.

PSO2: Develop computer-based systems to provide solutions for organizational, societal problems by working in multidisciplinary teams and pursue a career in the IT industry.



Program Outcomes (POs):

Engineering Graduates will be able to:

- **PO1. Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- **PO2. Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- **PO3. Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- **PO4. Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- **PO5. Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- **PO6. The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- **PO7. Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- **PO8. Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- **PO9. Individual and teamwork:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- **PO10. Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- **PO11. Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- **PO12. Life-long learning:** Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.



Course Objectives

1	To provide knowledge of software engineering discipline.
2	To make understand concept of planning, analyzing and configuration management
3	To make understand software engineering tools for project scheduling.
4	To introduce the concept of design & testing strategies to achieve good quality software
5	To introduce the concept of advance software methodology for building high quality applications
6	To make aware of computer-aided software engineering (CASE) tools.

Course Outcomes

At the end of the course student will be able to:		PO	Bloom Level
CSL501.1	Identify requirements and apply software process model to a given case study.	Apply	Apply (Level 3)
CSL501.2	Design Data Flow Diagrams for given case study.	Design	Evaluate (Level 5)
CSL501.3	Use software engineering tools for project scheduling and preparation of WBS.	Use	Apply (Level 3)
CSL501.4	Develop architectural models for the selected case study.	Develop	Create (Level 6)
CSL501.5	Develop test cases for the given case study on white box testing.	Develop	Create (Level 6)
CSL501.6	Use computer-aided software engineering (CASE) tools.	Use	Apply (Level 3)



Mapping of Experiments with Course Outcomes

Sr.No	Experiment	Course Outcomes					
		CSL501.1	CSL501.2	CSL501.3	CSL501.4	CSL501.5	CSL501.6
1	Apply Waterfall Model for a case study	3	--	--	--	--	--
2	Apply Spiral Model for a case study	3	--	--	--	--	--
3	Design DFD for a case study	--	3	--	--	--	--
4	To develop Software requirement Specification (SRS) in IEEE format for project	--	3	--	--	--	--
5	Prepare GANTT chart for a case study	--	--	3	--	--	--
6	Prepare WBS for a case study	--	--	3	--	--	--
7	Apply COCOMO Model for cost estimation	--	--	3	--	--	--
8	Apply FP based estimation technique for a case study	--	--	3	--	--	--
9	Develop architectural model for the selected case study	--	--	--	3	--	--
10	Develop test cases for White Box Testing for a given	--	--	--	--	3	--



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	code						
11	Preparation of Risk information sheet including RMMM plan	--	--	--	--	--	3

Enter correlation level 1, 2 or 3 as defined below

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation put “—”.



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Sr. No.	Name of Experiment	D.O.P.	D.O.C.	Page No.	Remark
1	Apply Waterfall Model for a case study				
2	Apply Spiral Model for a case study				
3	Apply RAD Model for a case study				
4	Design DFD for a case study				
5	To develop Software requirement Specification (SRS) in IEEE format for project				
6	Prepare GANTT chart for a case study				
7	Apply COCOMO Model for cost estimation				
8	Apply FP based estimation technique for a case study				
9	Develop architectural model for the selected case study				
10	Develop test cases for White Box Testing for a given code				
11	Preparation of Risk information sheet including RMMM plan				

D.O.P: Date of performance

D.O.C : Date of correction